STAT 429: Introduction to Time Series
Week 8 Lecture Notes

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1 Four Equivalent Representations of the Spectrum Model

We start by reviewing the spectrum model from four equivalent points of view. The fourth, in terms of random amplitudes and random phases, is perhaps the most natural way to understand the model: a time series is a superposition of sinusoids of different frequencies, each having a random amplitude and a random phase.

Consider observations $y_0, \ldots, y_{n-1}$. For simplicity, suppose n is odd and write $m = (n-1)/2$. Define the discrete Fourier transform (DFT) by

$$b_j = \sum_{t=0}^{n-1} y_t \exp\left(-\frac{2\pi i j t}{n}\right), \qquad j = 0, \ldots, n-1, \tag{1}$$

and the periodogram ordinates by

$$I(j/n) = \frac{|b_j|^2}{n}, \qquad j = 1, \ldots, m. \tag{2}$$

Since the observations are real, $b_{n-j} = \overline{b_j}$. It is therefore enough to model $b_1, \ldots, b_m$; the zero-frequency coefficient $b_0 = n\bar{y}$ is treated separately.

1.1 First representation: Gaussian DFT coefficients

The spectrum model can be defined by assuming that all the real and imaginary DFT coefficients are independent, with

$$\mathrm{Re}(b_j), \mathrm{Im}(b_j) \overset{\text{i.i.d.}}{\sim} N(0, \gamma_j^2), \qquad j = 1, \ldots, m. \tag{3}$$

The variance γ_j^2 is allowed to depend on the frequency j/n. Larger values of γ_j^2 mean that the corresponding Fourier component tends to have larger magnitude.

1.2 Second representation: sum of sines and cosines with Gaussian coefficients

The inverse DFT formula is

$$y_t = \frac{1}{n} \sum_{j=0}^{n-1} b_j \exp\left(\frac{2\pi i j t}{n}\right).$$

Pairing the terms at frequencies j/n and $(n-j)/n$ gives the real-valued representation

$$y_t = \beta_0 + \sum_{j=1}^{m} \left\{ \beta_{1j} \cos\left(\frac{2\pi j t}{n}\right) + \beta_{2j} \sin\left(\frac{2\pi j t}{n}\right) \right\}, \qquad t = 0, \ldots, n-1, \tag{4}$$

where

$$\beta_0 = \frac{b_0}{n}, \qquad \beta_{1j} = \frac{2\,\mathrm{Re}(b_j)}{n}, \qquad \beta_{2j} = -\frac{2\,\mathrm{Im}(b_j)}{n}. \tag{5}$$

Thus, equivalently, all the sine and cosine coefficients are independent and

$$\beta_{1j}, \beta_{2j} \overset{\text{i.i.d.}}{\sim} N(0, \tau_j^2), \qquad \boxed{\tau_j^2 = \frac{4\gamma_j^2}{n^2}}. \tag{6}$$

1.3 Third representation: exponential model for the periodogram

By (2) and (3),

$$I(j/n) = \frac{\mathrm{Re}(b_j)^2 + \mathrm{Im}(b_j)^2}{n} \overset{\text{ind.}}{\sim} \frac{\gamma_j^2}{n}\chi_2^2.$$

Because $\chi_2^2/2 \sim \mathrm{Exp}(1)$, where $\mathrm{Exp}(1)$ denotes the exponential distribution with mean one, we can equivalently write

$$\boxed{I(j/n) = f(j/n)\eta_j, \qquad \eta_j \overset{\text{i.i.d.}}{\sim} \mathrm{Exp}(1),} \tag{7}$$

with

$$\boxed{f(j/n) = \mathbb{E}I(j/n) = \frac{2\gamma_j^2}{n} = \frac{n\tau_j^2}{2}.} \tag{8}$$

The function f describes the average power at each frequency. This is a multiplicative-noise model: $\log I(j/n) = \log f(j/n) + \log \eta_j$. Note that $\log \eta_j$ does not have mean zero; $f(j/n)$ is the mean of the periodogram, not the exponential of the mean log-periodogram.

1.4 Fourth representation: a sum of sinusoids with random amplitudes and phases

We now rewrite each term in (4) as a single sinusoid. Define

$$R_j = \sqrt{\beta_{1j}^2 + \beta_{2j}^2}, \qquad \phi_j = \mathrm{atan2}(\beta_{2j}, \beta_{1j}) \pmod{2\pi}. \tag{9}$$

Then $\beta_{1j} = R_j \cos\phi_j$ and $\beta_{2j} = R_j \sin\phi_j$. Using $\cos(a-b) = \cos a \cos b + \sin a \sin b$, we get

$$\boxed{y_t = \beta_0 + \sum_{j=1}^{m} R_j \cos\left(\frac{2\pi j t}{n} - \phi_j\right).} \tag{10}$$

Why is this an equivalent probabilistic model? The pair (β_{1j}, β_{2j}) has an isotropic bivariate normal distribution. In polar coordinates its angle is uniform and independent of its radius. Consequently,

$$\boxed{\phi_j \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}(0, 2\pi),} \tag{11}$$

while

$$\frac{R_j^2}{\tau_j^2} = \frac{\beta_{1j}^2 + \beta_{2j}^2}{\tau_j^2} \sim \chi_2^2.$$

Therefore

$$\boxed{R_j^2 \overset{\text{ind.}}{\sim} \mathrm{Exp}(\text{mean } \lambda_j), \qquad \lambda_j = 2\tau_j^2.} \tag{12}$$

All amplitudes and phases are mutually independent across frequencies. The phases are identically distributed, but the amplitudes need not be: different frequencies have different λ_j.

Important distinction. It is R_j^2, not R_j, that is exponentially distributed. Equivalently, R_j has a Rayleigh distribution with scale τ_j, whose density is

$$p_{R_j}(r) = \frac{r}{\tau_j^2} \exp\left(-\frac{r^2}{2\tau_j^2}\right), \qquad r > 0.$$

This is a natural way to think about the spectrum model. We superpose sinusoids with deterministic frequencies j/n, but independently randomized phases and amplitudes. The parameter $\lambda_j = \mathbb{E}R_j^2$ tells us how much squared amplitude, on average, the jth sinusoid carries. A frequency with larger λ_j typically contributes a stronger oscillation.

1.5 Connecting all four sets of parameters

In the amplitude-phase representation, the DFT relations (5) imply

$$|b_j|^2 = \frac{n^2}{4}(\beta_{1j}^2 + \beta_{2j}^2) = \frac{n^2 R_j^2}{4}.$$

Hence the periodogram can be written directly in terms of squared amplitude:

$$\boxed{I(j/n) = \frac{nR_j^2}{4}.} \tag{13}$$

Taking expectations in (13),

$$f(j/n) = \mathbb{E}I(j/n) = \frac{n}{4}\mathbb{E}R_j^2 = \frac{n\lambda_j}{4}.$$

Thus, for every $j = 1, \ldots, m$, the four parameterizations are related by

$$\boxed{\gamma_j^2 = \frac{n^2\tau_j^2}{4} = \frac{n^2\lambda_j}{8} = \frac{nf(j/n)}{2},} \tag{14}$$

$$\boxed{\lambda_j = 2\tau_j^2 = \frac{8\gamma_j^2}{n^2} = \frac{4f(j/n)}{n},} \tag{15}$$

and

$$\boxed{f(j/n) = \frac{2\gamma_j^2}{n} = \frac{n\tau_j^2}{2} = \frac{n\lambda_j}{4}.} \tag{16}$$

These are not four different statistical models; they are four descriptions of the same model. We may work with whichever description is most convenient: the amplitude-phase representation for intuition, the Gaussian DFT representation for constructing the model, and the exponential periodogram representation for estimating its spectrum.

1.6 The case of even n

We assumed that n is odd (and $m = (n-1)/2$). If n is even, then $1/2$ becomes a Fourier frequency and $b_{n/2}$ becomes real (because $\sin(\pi t) = 0$ for all t). In this case, we can simply avoid working with $1/2$ by taking $m = (n-2)/2$ and using the model:

$$\mathrm{Re}(b_j), \mathrm{Im}(b_j) \overset{\text{i.i.d}}{\sim} N(0, \gamma_j^2) \quad \text{for } j = 1, \ldots, m.$$

This will be equivalent to (4). Basically everything will stay the same as before (only difference is that $m = (n-2)/2$). Here we are essentially forcing $\gamma_{n/2} = 0$. One can try to also try to estimate $\gamma_{n/2}$ using $b_{n/2} \sim N(0, \gamma_{n/2}^2)$ but this approach is slightly more complicated.

2 Regularized Estimation in the Spectrum Model

We now turn to parameter estimation. A key assumption is that the spectral power $f(j/n)$ should vary reasonably smoothly with frequency. Since there is an unknown parameter at each frequency, maximizing likelihood without regularization would lead to an estimate that simply reproduces the noisy periodogram.

2.1 Likelihood

We can write the likelihood using any of the equivalent representations. Under the Gaussian DFT model, the real and imaginary coefficients are independent normals with common variance γ_j^2 at frequency j/n. Ignoring factors that do not depend on the parameters, the likelihood is

$$\begin{aligned} L(\gamma_1, \ldots, \gamma_m) &\propto \prod_{j=1}^{m} \frac{1}{\gamma_j} \exp\left(-\frac{\mathrm{Re}(b_j)^2}{2\gamma_j^2}\right) \frac{1}{\gamma_j} \exp\left(-\frac{\mathrm{Im}(b_j)^2}{2\gamma_j^2}\right) \\ &= \prod_{j=1}^{m} \frac{1}{\gamma_j^2} \exp\left(-\frac{|b_j|^2}{2\gamma_j^2}\right) \\ &= \prod_{j=1}^{m} \frac{1}{\gamma_j^2} \exp\left(-\frac{nI(j/n)}{2\gamma_j^2}\right). \end{aligned}$$

Alternatively, from the exponential periodogram model (7), each $I(j/n)$ has an exponential distribution with mean $f(j/n)$, and hence the likelihood is

$$L(f) \propto \prod_{j=1}^{m} \frac{1}{f(j/n)} \exp\left(-\frac{I(j/n)}{f(j/n)}\right). \tag{17}$$

These are the same likelihood up to a multiplicative constant independent of the unknown parameters, because $f(j/n) = 2\gamma_j^2/n$.

2.2 Negative log-likelihood and overfitting

The negative log-likelihood in the DFT parameterization, after dropping constants, equals

$$\sum_{j=1}^{m} \left(2\log\gamma_j + \frac{nI(j/n)}{2\gamma_j^2}\right). \tag{18}$$

For optimization, it is convenient to work with the logarithmic parameters

$$\alpha_j = \log\gamma_j, \qquad j = 1, \ldots, m.$$

Then (18) becomes

$$\ell(\alpha) = \sum_{j=1}^{m} \left(2\alpha_j + \frac{nI(j/n)}{2} e^{-2\alpha_j}\right). \tag{19}$$

Without regularization, the minimization separates across frequencies. Differentiating the jth term gives

$$2 - nI(j/n)e^{-2\alpha_j} = 0.$$

Thus the unregularized estimates satisfy

$$\hat{\alpha}_j = \log\sqrt{\frac{nI(j/n)}{2}}, \qquad \hat{\gamma}_j^2 = \frac{nI(j/n)}{2}, \tag{20}$$

and therefore

$$\hat{f}(j/n) = \frac{2\hat{\gamma}_j^2}{n} = I(j/n).$$

In other words, the fitted spectrum exactly interpolates the periodogram. This is the overfitting that motivates regularization. (The displayed finite solution assumes $I(j/n) > 0$.)

2.3 Regularizing the log-scale parameters

If α_j varies smoothly with j, then the successive slopes $\alpha_{j+1} - \alpha_j$ should not vary too rapidly. This suggests penalizing the second differences

$$\Delta^2 \alpha_j = (\alpha_{j+1} - \alpha_j) - (\alpha_j - \alpha_{j-1}) = \alpha_{j+1} - 2\alpha_j + \alpha_{j-1}, \qquad j = 2, \ldots, m-1.$$

The ridge-type regularized estimator minimizes

$$\boxed{\hat{\alpha}^{\text{ridge}}(\lambda) = \underset{\alpha \in \mathbb{R}^m}{\arg\min} \left\{ \sum_{j=1}^{m} \left(2\alpha_j + \frac{nI(j/n)}{2} e^{-2\alpha_j} \right) + \lambda \sum_{j=2}^{m-1} (\Delta^2 \alpha_j)^2 \right\}.} \tag{21}$$

The lasso-type regularized estimator minimizes

$$\boxed{\hat{\alpha}^{\text{lasso}}(\lambda) = \underset{\alpha \in \mathbb{R}^m}{\arg\min} \left\{ \sum_{j=1}^{m} \left(2\alpha_j + \frac{nI(j/n)}{2} e^{-2\alpha_j} \right) + \lambda \sum_{j=2}^{m-1} |\Delta^2 \alpha_j| \right\}.} \tag{22}$$

The first penalty encourages gradual changes in slopes, whereas the second encourages many second differences to be zero, resulting in stretches where α_j is approximately linear in j. Both objectives are convex in α.

Once we have estimated α_j, we obtain estimates in each equivalent parameterization by the transformations

$$\hat{\gamma}_j = e^{\hat{\alpha}_j}, \qquad \hat{\tau}_j^2 = \frac{4}{n^2} e^{2\hat{\alpha}_j}, \qquad \hat{\lambda}_j = \frac{8}{n^2} e^{2\hat{\alpha}_j}, \qquad \boxed{\hat{f}(j/n) = \frac{2}{n} e^{2\hat{\alpha}_j}.} \tag{23}$$

The resulting estimated spectrum is smoother and more interpretable than the raw periodogram, while still allowing the strength of the oscillations to vary with frequency.

2.4 Some Questions and Final Remarks

We conclude our discussion of spectral analysis with three questions.

1. Suppose a monthly time series has a prominent spectral peak at frequency 0.025. What is the corresponding period?

   The period is the reciprocal of the frequency:

   $$T = \frac{1}{0.025} = 40 \text{ months}.$$

   Thus, the time series has a strong oscillatory component with a period of approximately 40 months.

2. Suppose a time series has a strong spectral peak corresponding to a period of four years. Does this mean that the pattern repeats exactly every four years?

   No. A spectral peak at this frequency means that oscillations with a period of approximately four years contribute substantially to the variability of the time series. It does not imply that the observations repeat exactly every four years, nor does it imply that we can predict the timing of future events precisely.

3. Why do we need to smooth or regularize the periodogram?

   Recall that under our spectrum model,

$$I(j/n) = f(j/n)\eta_j, \qquad \eta_j \overset{\text{i.i.d.}}{\sim} \text{Exp}(1).$$

   Consequently,

$$\mathbb{E}[I(j/n)] = f(j/n), \qquad \text{Var}(I(j/n)) = f(j/n)^2.$$

   Thus, the standard deviation of each periodogram ordinate equals its mean. The raw periodogram is therefore quite noisy, even for a long time series.

   Regularization allows us to borrow information from neighboring frequencies to obtain a smoother and more informative estimate of the underlying spectrum.

Final Remark. The most important point to remember is that *the spectrum describes the relative strength of oscillations at different frequencies. It does not assert that the time series is exactly periodic.*

The examples considered in the code illustrate several different uses of spectral analysis. Sunspot data exhibit a strong approximately periodic cycle. The Southern Oscillation Index illustrates irregular recurrence, with substantial power at frequencies corresponding to multiyear oscillations. Earthquake vibration data show how spectral analysis can identify which frequencies are suppressed by a control system. Industrial production data illustrate how the spectrum can reveal characteristic timescales of economic fluctuations.

In all these examples, the spectrum provides a useful description of the time series even when the underlying oscillations are neither perfectly regular nor deterministic.

3 Auto or Lagged Regression

Our next topic of study is AutoRegression. The main models here are called ARIMA. ARIMA is an acronym standing for Auto-Regressive Integrated Moving Average. We will first study Auto-Regressive (AR) models, then we shall include the MA part to get ARMA models, finally we see what "Integrated" means.

3.1 AR (Auto-Regressive) Models

We observe time series $y_1, \ldots, y_n$. AR models are simply linear regression models where the covariates are chosen to be past (or lagged) values of y_t.

The AR model of order p (referred to by $AR(p)$) is given by

$$y_t = \phi_0 + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \epsilon_t \tag{24}$$

for $t = p+1, \ldots, n$. In matrix notation,

$$Y = X\beta + \epsilon$$

where

$$Y = \begin{pmatrix} y_{p+1} \\ y_{p+2} \\ \cdot \\ \cdot \\ \cdot \\ y_n \end{pmatrix} \quad X = \begin{pmatrix} 1 & y_p & y_{p-1} & \cdot & \cdot & \cdot & y_1 \\ 1 & y_{p+1} & y_p & \cdot & \cdot & \cdot & y_2 \\ 1 & y_{p+2} & y_{p+1} & \cdot & \cdot & \cdot & y_3 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & y_{n-1} & y_{n-2} & \cdot & \cdot & \cdot & y_{n-p} \end{pmatrix} \quad \beta = \begin{pmatrix} \phi_0 \\ \phi_1 \\ \phi_2 \\ \cdot \\ \cdot \\ \cdot \\ \phi_p \end{pmatrix} \quad \epsilon = \begin{pmatrix} \epsilon_{p+1} \\ \epsilon_{p+2} \\ \cdot \\ \cdot \\ \cdot \\ \epsilon_n \end{pmatrix}$$

This regression model is called AutoRegression because the responses as well as the covariates are both formed from the same time series: the time series y_t is regressed on its own lagged values $y_{t-1}, \ldots, y_{t-p}$.

The parameters $\phi_0, \ldots, \phi_p$ are estimated in the usual way by minimizing $\|Y - X\beta\|^2$. Let the estimates be $\hat{\phi}_0, \ldots, \hat{\phi}_p$.

AR models are useful for predicting future values of the time series. For predicting y_{n+1}, we plug $t = n + 1$ in (24) to get

$$\hat{y}_{n+1} = \hat{\phi}_0 + \hat{\phi}_1 y_n + \hat{\phi}_2 y_{n-1} + \cdots + \hat{\phi}_p y_{n+1-p}.$$

Note that $y_n, y_{n-1}, \ldots, y_{n+1-p}$ are all observed and they are the last p observations. For predicting y_{n+2}, we plug $t = n + 2$ in (24) to get

$$\hat{y}_{n+2} = \hat{\phi}_0 + \hat{\phi}_1 y_{n+1} + \hat{\phi}_2 y_n + \cdots + \hat{\phi}_p y_{n+2-p}.$$

In the above, y_{n+1} is not observed. But we can replace it by the predicted value $\hat{y}_{n+1}$. This gives

$$\hat{y}_{n+2} = \hat{\phi}_0 + \hat{\phi}_1 \hat{y}_{n+1} + \hat{\phi}_2 y_n + \cdots + \hat{\phi}_p y_{n+2-p}.$$

More generally, we predict y_{n+i} by the recursion

$$\hat{y}_{n+i} = \hat{\phi}_0 + \hat{\phi}_1 \hat{y}_{n+i-1} + \cdots + \hat{\phi}_p \hat{y}_{n+i-p} \quad \text{for } i = 1, 2, \ldots$$

where the recursion is initialized with

$$\hat{y}_j = y_j \quad \text{for } j = n, n-1, \ldots, n+1-p.$$

We will study AR models in more detail later. Here, we provide a motivation for their use through the sunspots dataset. This was how the AR models were originally invented by Yule (1927).

3.2 AR Models for the Sunspots Data

For the sunspots dataset, we previously employed the model

$$y_t = \beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t) + \epsilon_t \quad \text{for } t = 1, \ldots, n \tag{25}$$

We used a Bayesian method to infer the frequency parameter f (which is the main parameter of interest) and this led to an estimated period of close to 11 years (which is often cited as the period of the solar cycle). Note however that (25) is not ideal for the sunspots dataset for at least two reasons: (a) the fit to the data is not very good (some of the oscillations have a much higher amplitude than that explained by the single sinusoid), (b) data generated from the model (25) look much more "noisy" compared to the actual sunspots data. Starting with these observations, Yule (1927) proposed an alternative model that is also based on a single sinusoid. This alternative model is based on the idea of AR modeling.

Yule started with the following basic observation. Let s_t denote the sinusoid:

$$s_t = \beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t) \tag{26}$$

The same sinusoid can be understood as the solution to a specific *difference equation*. To derive the difference equation, let us first note that, in continuous time, $s(t)$ satisfies

$$s''(t) = -(2\pi f)^2 \left(\beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t)\right) = -(2\pi f)^2 \left(s(t) - \beta_0\right). \tag{27}$$

In discrete time (where $t \in \{\ldots, -2, -1, 0, 1, 2, \ldots\}$), the sequence (26) satisfies the following difference equation that is analogous to (27):

$$s_t - 2s_{t-1} + s_{t-2} = 2(\cos(2\pi f) - 1)\left(s_{t-1} - \beta_0\right). \tag{28}$$

To see this, note that (below we take $\omega = 2\pi f$ for notational simplicity)

$$\begin{aligned} s_t - 2s_{t-1} + s_{t-2} &= \beta_1 \left(\cos(\omega t) - 2\cos(\omega(t-1)) + \cos(\omega(t-2))\right) \\ &\quad + \beta_2 \left(\sin(\omega t) - 2\sin(\omega(t-1)) + \sin(\omega(t-2))\right) \end{aligned}$$

Writing $A = \omega(t-1)$ and $B = \omega$, we get

$$\begin{aligned} \cos(\omega t) - 2\cos(\omega(t-1)) + \cos(\omega(t-2)) &= \cos(A+B) - 2\cos A + \cos(A-B) \\ &= 2\cos A(\cos B - 1) \\ &= 2(\cos\omega - 1)\cos(\omega(t-1)) \end{aligned}$$

and similarly

$$\sin(\omega t) - 2\sin(\omega(t-1)) + \sin(\omega(t-2)) = 2(\cos\omega - 1)\sin(\omega(t-1)).$$

This proves

$$\begin{aligned} s_t - 2s_{t-1} + s_{t-2} &= 2(\cos\omega - 1)\left(\beta_1\cos(\omega(t-1)) + \beta_2\sin(\omega(t-1))\right) \\ &= 2(\cos\omega - 1)(s_{t-1} - \beta_0) \end{aligned}$$

thereby establishing (28).

The converse is also true in the sense that every solution $\{s_t\}$ to the difference equation (28) say, for $t = 3, 4, 5, \ldots$, with given values of s_1 and s_2 (initial conditions) is of the form (26) for some β_1 and β_2. To see this, let $g_t = s_t - \beta_0$ and note that $\{g_t\}$ satisfies

$$g_t - 2g_{t-1} + g_{t-2} = 2(\cos\omega - 1)g_{t-1}.$$

For $0 < f < 1/2$, we find β_1 and β_2 such that (note again that $\omega = 2\pi f$)

$$h_t := \beta_1\cos(\omega t) + \beta_2\sin(\omega t)$$

matches g_t for $t = 1, 2$. Now if $g_{t-1} = h_{t-1}$ and $g_{t-2} = h_{t-2}$, then

$$\begin{aligned} g_t &= (2\cos\omega)g_{t-1} - g_{t-2} \\ &= (2\cos\omega)h_{t-1} - h_{t-2} \\ &= (2\cos\omega)\left(\beta_1\cos(\omega(t-1)) + \beta_2\sin(\omega(t-1))\right) - \left(\beta_1\cos(\omega(t-2)) + \beta_2\sin(\omega(t-2))\right) \\ &= \beta_1\left(2\cos\omega\cos(\omega(t-1)) - \cos(\omega(t-2))\right) + \beta_2\left(2\cos\omega\sin(\omega(t-1)) - \sin(\omega(t-2))\right). \end{aligned}$$

Verify that

$$2\cos\omega\cos(\omega(t-1)) - \cos(\omega(t-2)) = \cos(\omega t)$$

and

$$2\cos\omega\sin(\omega(t-1)) - \sin(\omega(t-2)) = \sin(\omega t),$$

which gives

$$g_t = \beta_1\cos(\omega t) + \beta_2\sin(\omega t) = h_t.$$

We thus proved that if $g_{t-1} = h_{t-1}$ and $g_{t-2} = h_{t-2}$, then $g_t = h_t$. Using this for $t = 3, 4, \ldots$ proves that (26) is the unique solution to (28).

To summarize, an alternative way of describing a sinusoid of frequency $\omega = 2\pi f$ is via the difference equation (28) which is equivalent to

$$s_t = (2\cos\omega)s_{t-1} - s_{t-2} + 2(1-\cos\omega)\beta_0.$$

Based on this equation, Yule proposed the model:

$$y_t = \phi_0 + \phi_1 y_{t-1} - y_{t-2} + \epsilon_t \tag{29}$$

with two parameters ϕ_0 and ϕ_1 (and the additional noise parameter σ in $\epsilon_t \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$). Note that (29) is also a single sinusoid plus noise model but now the noise is in a different place.

To better understand the difference between (29) and the earlier model (25), consider the following physical situation where sinusoids naturally arise (see e.g., page 2 of the Fourier Analysis book by Stein and Shakarchi). Consider a mass m that is attached to a horizontal spring, which itself is attached to fixed wall, and assume that the system lies on a frictionless surface. Suppose that β_0 is the location of the center of the mass when the spring is neither compressed or stretched. When the spring is compressed or stretched and released, the mass undergoes simple harmonic motion.

Let $s(t)$ denote the position of the mass at time t. Hooke's law says that the force exerted by the spring on the mass is given by $F = -\kappa\,(s(t) - \beta_0)$ where $\kappa > 0$ is the spring constant. By Newton's law (note that the acceleration is given by $s''(t)$), we have

$$-\kappa\,(s(t) - \beta_0) = m s''(t)$$

This is same as

$$s''(t) = -\omega^2\,(s(t) - \beta_0) \quad \text{where } \omega := \sqrt{\frac{\kappa}{m}}$$

whose general solution is the sinusoid $s(t) = \beta_0 + \beta_1 \cos(\omega t) + \beta_2 \sin(\omega t)$. In the context of this physical situation, the two different noisy sinusoid models ((25) and (29)) can be understood as follows. We are taking measurements of the displacement y_t at various times t.

Model (25): Here our measurements are noisy and every measurement is corrupted by an unknown noise which we are terming ϵ_t and modeling as $N(0, \sigma^2)$.

Model (29): Here there is no measurement error and our measurement mechanism is perfect. However the actual oscillation of the mass is not perfectly sinusoidal and is affected by noise. For example, imagine, as Yule put it, that some kids are randomly throwing stones at the mass (sometimes from the left and sometimes from the right) while it is oscillating.

It is very interesting to note that observations generated from Model (29) are much smoother compared to observations generated from Model (25). Yule used this to argue that (29) is a better model for the sunspots data compared to (25).

The AR(2) model is:

$$y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t \tag{30}$$

(29) can be seen as a simpler version of the above model where the ϕ_2 parameter is set to the value -1. Yule fit both models ((29) and (30)) to the sunspots dataset. It is interesting that these two models give different predictions for future sunspots values: (29) gives sinusoidal predictions while (30) gives *damped* sinusoidal predictions. We shall discuss later how different parameter settings for AR models lead to different predictions.

A very nice account of Yule's influential 1927 paper is Chapter 6 of the 2011 book "The Foundations of Modern Time Series Analysis" by T. C. Mills. (available for free from the library website). Yule's 1927 paper itself is available freely online.